RQ4 -- Counterfactual Reasoning and Algorithmic Recourse

Module AAA (template run) | OULAD Causal AI Project | Prepared for review

1. Objective

RQ3 identified why a student's outcome was anomalous. RQ4 asks two forward-looking questions: (1) what would a student's outcome have been had something been different (**counterfactual reasoning**), and (2) what concrete, feasible change would move a student to Pass or higher (**algorithmic recourse**). Recourse is generated with DiCE-ML, a classifier-based method, and then verified through the fitted Structural Causal Model (SCM) from RQ3 to test whether it would actually work causally. This closes the gap between associational recourse (what a classifier says will work) and causal recourse (what the fitted causal model says will actually work).

2. Scope of this run

Structure	Nodes retained	Status
DAG1 -- Demographics	studied_credits -> final_result (2 nodes)	Exploratory only -- weak fit, invertibility rejected
DAG2 -- Assessments	ass_1 ... ass_5, final_result (6 nodes)	Full pipeline -- strongest SCM in AAA
DAG3 -- VLE Clicks	clicks_resource -> final_result (2 nodes)	Full pipeline -- Option B (see Sec. 4)
Combined DAG	gender, highest_education, studied_credits, age_band, total_clicks, ass_1...ass_5, final_result (11 nodes)	Full pipeline -- main result

Cohorts: Withdrawn (final_result = 0, n = 126), Fail (final_result = 1, n = 91), Success (final_result in {2, 3}, n = 531, reference group only). All cohorts exceed the n = 30 flag threshold, so no results are sample-size caveated in AAA.

3. Key decisions locked for this run

Decision	Value	Rationale
Success threshold	final_result >= 1.5 (2.0 reported as sensitivity check)	Midpoint between Fail and Pass on continuous SCM output
DiCE method	random, 3 candidates per student	Balances speed with enough candidates for a diversity metric
studied_credits	Treated as immutable	Fixed at enrollment; cannot be changed after the fact
Checkpoint constraint	Deferred	Layers 1-2 run end-to-end first; added later if time allows
Intervention styles	Both reported: shift to Success-group median, and a bounded step nudge (+10 pts / +1 engagement bin)	Median asks how far a student is from a typical successful one; the step asks how responsive the outcome is to a realistic nudge

4. SCM readiness and a methodological correction

RQ3's saved SCMs were plain *StructuralCausalModel* objects, which do not support the noise recovery counterfactual reasoning requires. Confirmed by testing (not assumed): none of the RQ3 pickles were usable. All four SCMs were refit as *InvertibleStructuralCausalModel*, using the same edge lists, exclusions, and overrides as RQ3.

New finding not visible in RQ3: the combined SCM's total_clicks mechanism initially failed its invertibility check (p = 0.003) -- a problem RQ3's plain SCM never tested for. Fixed with the same LassoCV override already used for age_band (R2 improved 0.059 -> 0.098, p rose to 0.23, not rejected). This override is now standard practice and will be checked for every module going forward, even where RQ3 did not flag total_clicks as a concern.

final_result (a 4-level ordinal outcome modelled with continuous noise) fails its invertibility check in all four SCMs. This is expected and is handled by reporting results at both the 1.5 and 2.0 thresholds; documented as a standing limitation rather than treated as an error.

5. Layer 1 -- Counterfactual reasoning

5.1 Shift-to-Success results

Node (DAG)	Withdrawn -> Pass	Fail -> Pass	Note
ass_4_score (DAG2)	88.9%	63.7%	Strongest single lever in AAA
ass_2_score (DAG2)	23.0%	52.7%	
ass_1_score (DAG2)	27.0%	46.2%	
ass_5_score (DAG2)	47.6%	49.5%	
clicks_resource (DAG3)	17.5%	44.0%	Weakest; DAG3's fit is weak (R2=0.08)
total_clicks (Combined)	0.0%	52.7%	Insufficient alone for Withdrawn (see 5.2)

Cross-method consistency: ass_4_score is ranked the top driver by three independent methods -- RQ3's anomaly attribution (top Withdrawn contributor, mean 0.461), the standalone DAG2 counterfactual, and the Combined-DAG counterfactual (identical 88.9% result). This agreement across methods is a strong basis for treating ass_4_score as AAA's primary early-warning signal.

5.2 Cross-check against RQ2 effect sizes

Every shift-to-median result is 2-4x larger than the corresponding RQ2 average treatment effect (ATE). This is expected, not a discrepancy: the ATE is a per-unit linear effect, while shifting a student to the Success-group median is a much larger jump. total_clicks alone cannot move Withdrawn students to Pass (0% under either threshold or style) -- engagement alone is not sufficient without the assessment cascade also improving; for Fail students, already closer to Pass, total_clicks performs comparably to the assessments (about 53%).

5.3 Fairness counterfactuals

Flipping gender, highest_education, or studied_credits produced an exactly zero shift in outcome for every student (confirmed, not a rounding artifact). This is not a null fairness finding -- it reflects that all three variables reach final_result only through age_band, and age_band's mechanism is too weak (R2 = 0.03, invertibility still rejected) to carry the upstream signal through noise recovery. These three are reported as **not estimable** in AAA, a distinct claim from “no effect.” age_band intervened on directly does show a real effect (52.7% of Fail reach Pass). Before trusting a fairness counterfactual in later modules, the mediating node's own fit must be checked, not just whether a causal path exists.

6. Layer 2 -- Algorithmic recourse

DiCE-ML generates candidate changes that flip a trained classifier's prediction to “Success.” Each candidate is then pushed through the SCM as an intervention to test whether it would actually work causally. Two verification modes are used: **hard** (every changed feature is forced to DiCE's value) and **propagated** (only upstream changes are forced; downstream features are left to respond through the SCM, which is the stricter, more causally honest test).

6.1 Validity gap by pipeline

Pipeline	Classifier AUC	DiCE validity	Causal validity (hard)	Causal validity (prop.)
DAG2 -- Assessments	0.934	100%	48.7% / 56.0%*	21.7% / 60.1%*
DAG3 -- Clicks (Option B)	0.811	100%	17.5% / 19.8%*	17.5% / 19.8%* (identical)
Combined DAG	0.934	100%	49.2% / 50.5%*	26.5% / 56.4%*

* Withdrawn / Fail. DiCE validity is 100% by construction (it only returns candidates it believes succeed).

6.2 Headline finding: the validity gap depends on how far a student starts

The gap between DiCE validity and causal validity is not a fixed discount -- it depends on the student's starting distance from the target, and runs in **opposite directions** for the two cohorts (consistent in both DAG2 and the

Combined DAG):

• **Withdrawn** (starts at final_result = 0): propagated validity collapses relative to hard (DAG2: 48.7% -> 21.7%; Combined: 49.2% -> 26.5%). Most two-feature recourse candidates work only because DiCE hardcodes a second assessment score that the SCM, left to generate it from the student's own weak-performer profile, would not independently produce.

• **Fail** (starts at final_result = 1, one level below Pass): propagated validity is slightly *higher* than hard (DAG2: 56.0% -> 60.1%; Combined: 50.5% -> 56.4%). These students are close enough to the target that the SCM's own generated value for a dropped feature performs at least as well as DiCE's hardcoded one.

6.3 DAG3 (Option B): the clearest demonstration of the validity gap

DAG3's SCM recognises only clicks_resource as causally connected to the outcome. DiCE was trained and run on **all nine click types** rather than restricted to clicks_resource, so that the classifier could recommend click types with no real causal effect -- exposing the gap directly. Result: causal validity is **17.5% (Withdrawn) / 19.8% (Fail)** against 100% DiCE validity -- roughly **80% of DiCE's "successful" click-based recourse would not actually move the student to Pass.** Only 42% of Withdrawn and 31% of Fail candidates even touch clicks_resource at all; the rest recommend changes with zero causal path to the outcome. This is the starkest validity gap of any pipeline in AAA and the clearest case for why causal verification is necessary.

6.4 A limitation of DiCE-driven recourse itself

In the Combined DAG, total_clicks is the only genuinely free-to-change (actionable) lever -- assessment scores are better read as achievable performance levels than something a student can directly will into being. Yet DiCE recommends changing total_clicks in only **6.6% of Withdrawn and 10.6% of Fail candidates**, overwhelmingly preferring to edit assessment scores directly. The selection-frequency chart in Section 7 shows this visually: total_clicks is the least-selected node of all six, despite being the one node a student can actually act on directly. This is reported as a limitation of the recourse method itself, worth checking in later modules where total_clicks has a stronger mechanism fit.

7. Visualizations

One representative, causally-valid (propagated mode) recourse example is shown per cohort, illustrating what the method recommends for one real student.

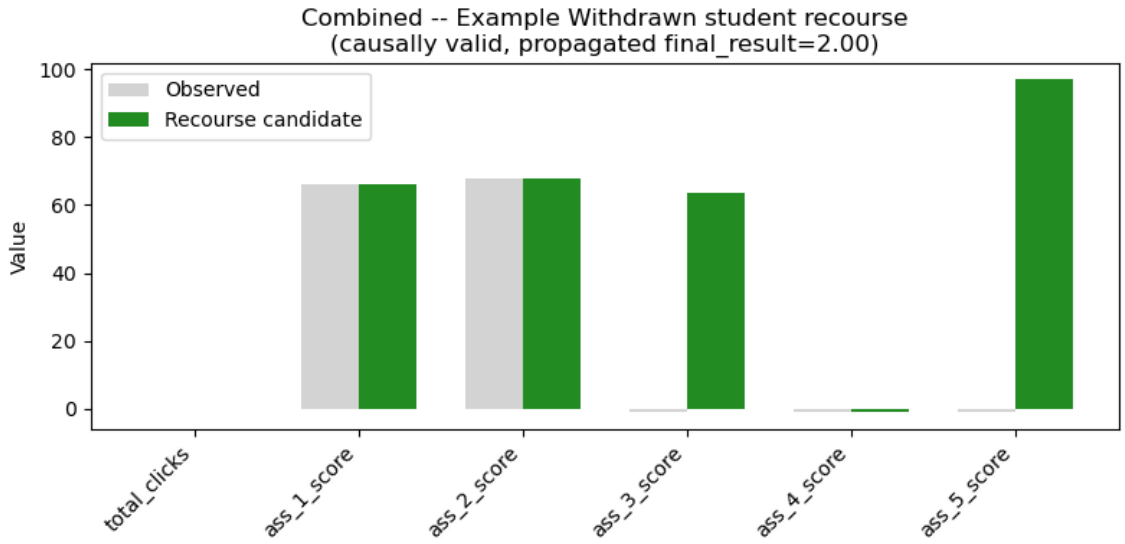


Figure 1. Example Withdrawn student -- observed vs. recourse candidate (Combined DAG, reaches final_result = 2.0)

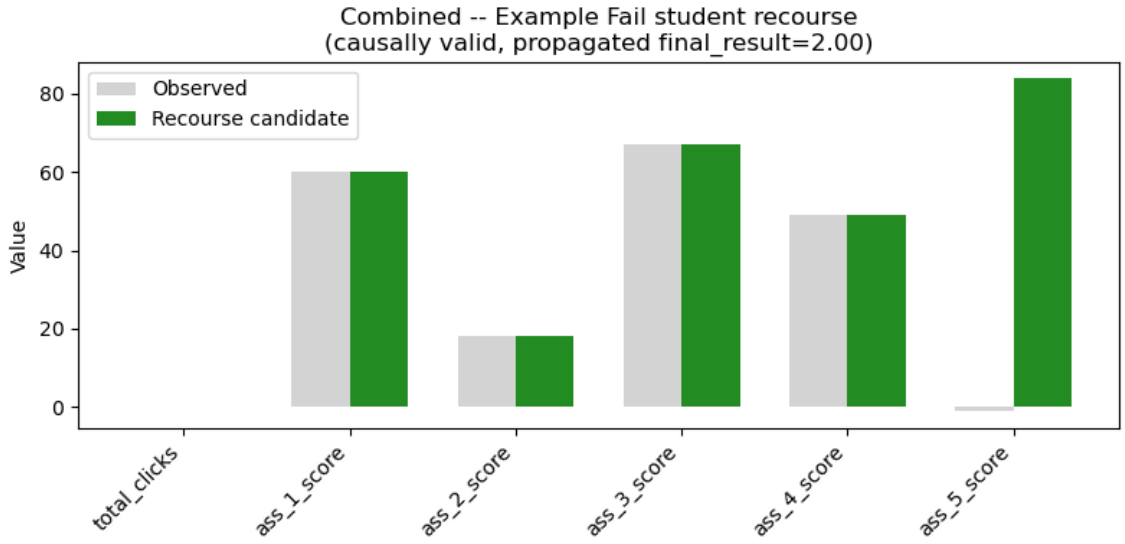


Figure 2. Example Fail student -- observed vs. recourse candidate (Combined DAG, reaches final_result = 2.0)

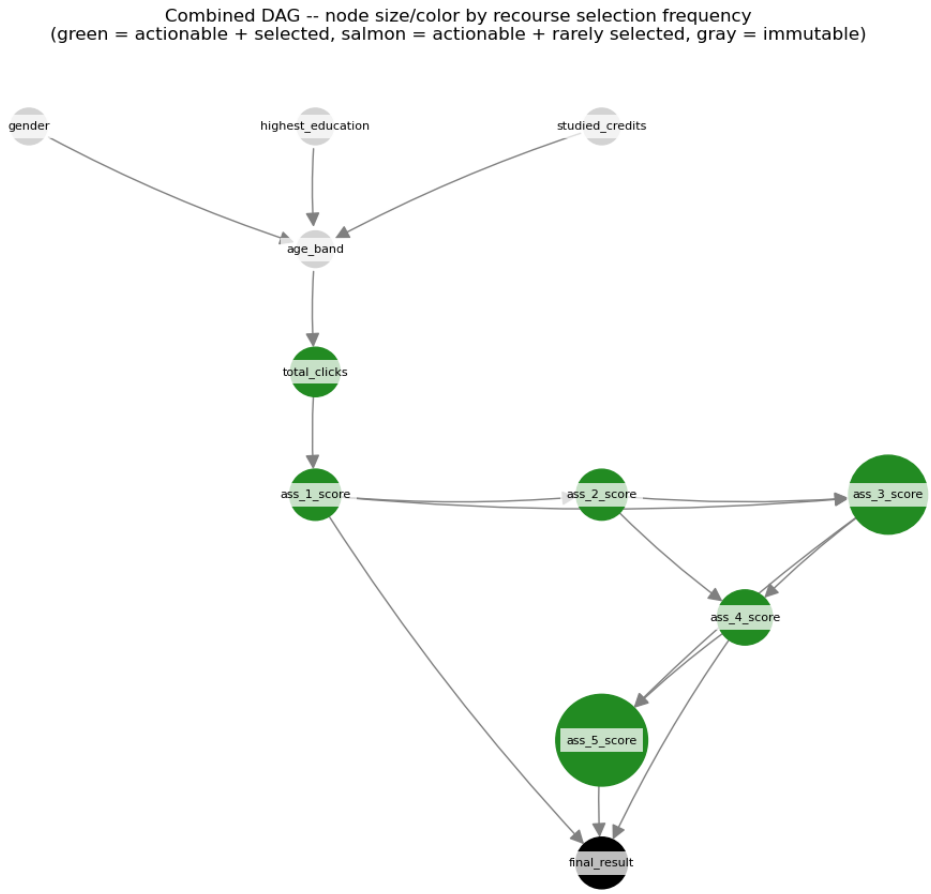


Figure 3. Combined DAG with nodes sized by recourse selection frequency across all 651 candidates. Green = actionable and selected (larger = more often); gray = immutable. Note total_clicks is the smallest selected node despite being the only freely actionable lever (see Section 6.4).

8. Limitations

• final_result invertibility is rejected in all four SCMs -- an expected property of a 4-level ordinal outcome modelled with continuous noise. Mitigated by reporting both 1.5 and 2.0 thresholds (results were identical in AAA; to be re-checked per module).

• Fairness counterfactuals on gender, highest_education, and studied_credits are not estimable in AAA -- all three route through a weakly-fit mediator (age_band).

• DAG1 and DAG3 are structurally thin (a single edge each after RQ3's reachability exclusions), so their standalone fit is weak; DAG1 is treated as exploratory only.

• AAA's recourse direction-dependent validity gap and the total_clicks under-recommendation pattern are observed in one module only -- both are flagged to be explicitly checked against BBB-GGG rather than assumed to generalize.

9. Status and next step

AAA is complete end-to-end: Step 0 (SCM readiness), Layer 1 (shift-to-success and fairness counterfactuals) and Layer 2 (recourse generation, causal verification, and metrics) for all four SCMs, with visualizations. AAA serves as the finalized template. Next step is to replicate the same pipeline to BBB through GGG, reusing every decision locked in Section 3 and the total_clicks invertibility check from Section 4 as a standard step for every module.